

NAME _____

CLASS 2T _____



Catholic Junior College
JC2 Preliminary Examinations
Higher 2

PHYSICS

9749/01

Multiple Choice Questions

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and tutorial group on this cover page.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet (OMR sheet), unless this has been done for you.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are a total of 30 Multiple Choice Questions in this paper.

Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Answer Sheet (OMR sheet) provided.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

PHYSICS DATA:

speed of light in free space	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$ $\approx (1/(36\pi)) \times 10^{-9} \text{ F m}^{-1}$
elementary charge	e	$= 1.60 \times 10^{-19} \text{ C}$
the Planck constant	h	$= 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass constant	u	$= 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron	m_e	$= 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton	m_p	$= 1.67 \times 10^{-27} \text{ kg}$
molar gas constant	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
the Avogadro constant	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant	k	$= 1.38 \times 10^{-23} \text{ mol}^{-1}$
gravitational constant	G	$= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall	g	$= 9.81 \text{ m s}^{-2}$

PHYSICS FORMULAE:

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on / by a gas	$W = p \Delta V$
hydrostatic pressure	$P = \rho gh$
gravitational potential	$\phi = -\frac{Gm}{r}$
temperature	$T / K = T / ^\circ C + 273.15$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
mean translational kinetic energy of an ideal gas molecule	$E = \frac{3}{2} kT$
displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $= \pm \omega \sqrt{x_0^2 - x^2}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \dots$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

- 1** A student measured the mass and linear dimensions of a rectangular block in order to determine its density. The measurements are:

Height = (1.00 ± 0.01) cm

Length = (5.00 ± 0.01) cm

Breadth = (2.00 ± 0.01) cm

Mass = (80.0 ± 0.2) g

What is the uncertainty in the value of the density calculated?

A 0.01 g cm^{-3}

B 0.02 g cm^{-3}

C 0.1 g cm^{-3}

D 0.2 g cm^{-3}

- 2** Express the *volt* in SI base units